How does perceived colour vibrancy (vibrant vs subtle) of a sports car influence its aesthetic appeal?

Research Question:

This study examines how sports car colour vibrancy (vibrant vs subtle) influences consumers' perceived aesthetic appeal. In automotive branding, colour is a key component that signals dynamism or restraint, shaping attractiveness and market positioning (Babu P and K. M, 2024). Drawing on visual perception (Purves, 2025) and signalling theory (Choudhury, 2024), vibrancy is conceptualised as an affective cue that shapes consumers' immediate emotional impressions of a sports car's appearance, thereby influencing perceived aesthetic appeal. This is important for marketers as colour can quickly shape first impressions and positioning.

Hypothesis:

This study examines whether sports car colour vibrancy influences perceived aesthetic appeal within an automotive evaluation context. Colour, also termed as colourfulness, saturation or vividness, refers to the intensity of colour, ranging from weak or pale to strong or deep (Cheung, 2021). Research in colour psychology suggests that saturated colours evoke stronger arousal and attention responses, shaping product evaluations (Labrecque and Milne, 2011). Aesthetics is a key factor when it comes to sports car brand perception and purchase interest (Corina Paula Tarţa et al., 2020).

Drawing on the signalling theory, observable product attributes such as aesthetics convey symbolic importance. Within performance-oriented categories such as sports cars, vibrant colours tend to indicate dynamism, boldness and excitement (Wheels & Motion Editorial, 2024). Heightened visual intensity may hence enhance affective evaluation and increase perceived attractiveness. These studies have inspired the following hypothesis:

H1: Sports cars in high-vibrancy colours will be perceived as significantly more aesthetically appealing than sports cars in low-vibrancy colours.

This hypothesis clearly specifies the independent variable (Colour Vibrancy) and perceived aesthetic appeal (dependent variable) within a clearly specified consumer evaluation setting.

Experiment Design:

This study employs two-condition between-subjects experimental design to examine how perceived colour vibrancy (vibrant vs subtle) of a sports car influences aesthetic appeal. Participants are randomly assigned to one of the two colour conditions exposed to a single visual stimulus. The stimulus consists of an image of a Porsche 911 displayed in a natural driving setting. Each participant sees only one version of the image as it makes it harder for them to detect the manipulation or compare versions directly (Ruhl, 2023).

The **Independent Variable** was colour vibrancy condition (subtle vs vibrant). Using the same base photograph, two versions are created through AI prompts. In the vibrant photo, the car's body colour is vivid red, which is so because of ~26% increased saturation. In the subtle

condition, the exact same photo is used but the saturation of the car's body colour was reduced to create a muted red tone. The stimuli were identical in car model, number plate, camera angle, background and lighting (with all other visual elements held constant), ensuring that vibrancy was the only difference. (Figure 1 and Figure 2 for reference)



Figure 1: Control Group Image.



Figure 2: Treatment Group Image.

Participants for this survey are online and are randomly assigned one of the two conditions in a 1:1 ratio using qualtrics' randomizer. No eligibility or screening was applied. Participants were instructed to view the image carefully before clicking Next to proceed. They then completed the

aesthetic appeal items, followed by manipulation check questions assessing perceived vibrancy. Finally, sports-car interest (control), an attention check, and demographics were collected.

Assigning randomly distributes individual differences evenly across conditions, reducing selection bias. By holding all non-colour attributes constant and exposing participants to only one stimulus, the design isolates the causal effect of colour vibrancy on perceived aesthetic appeal, thereby strengthening internal validity and ensuring the study is replicable.

Refer to Appendix for questionnaire.

Measurement:

All factors were measured using a structured online questionnaire administered immediately after exposure to the stimulus image. Measures were presented in the following order: Dependent Variable, Manipulation check, Control variables and Attention check and demographics. DV and MC were assessed using a 7-point Multi-item Likert-scale(1=Strongly disagree, 7=Strongly agree), with attention check using a 5-point Single-item Likert-scale and Interest being a 3-point bipolar semantic differential scale.

The **Dependent Variable** was perceived aesthetic appeal and was measured using three items: “This sports car is visually appealing”, “I find the design of this car aesthetically pleasing”, and “I like the overall look of this car.” These items capture visual liking, consistent with established definitions of aesthetic appeal as a subjective response to product design (Blijlevens et al., 2017). Perceived aesthetic appeal refers to an individual’s personal judgement of how visually attractive or pleasing a stimulus appears (Reppa and McDougall, 2022). Responses were averaged to form an overall aesthetic appeal score, with higher values indicating greater appeal.

The **Manipulation Check** assessed perceived colour vibrancy using two questions; "The colour of this car appears vivid" and "The colour of this car appears subtle." They were selected because they capture opposite ends of the vibrancy, allowing clear assessment of whether participants perceived the intended difference between conditions. The term “subtle” was reverse-coded and responses were averaged to form a composite perceived vibrancy score, with higher values indicating stronger perceptions of colour vividness.

Interest in sports cars was included as **Control Variable** measured with three semantic differential items (Unexcited-Excited; Indifferent-Enthusiastic; Uninterested-Very interested) Responses were averaged into a single involvement score, with higher values indicating greater interest in sports cars. This was included as individuals with greater enthusiasm may evaluate the car more positively regardless of colour condition.

Attention Check: "Select Somewhat Agree for this item"; other responses were coded as fail and excluded. **Age** and **Gender** were collected as standard demographic variables, given potential variation in aesthetic perception across groups.

Sample Size and Data:

Seventy survey responses were collected; however, an embedding-code error prevented data export, so an AI-generated dataset through ChatGPT5.2 was used (Prompt-Appendix 1). Participation was voluntary, with informed consent obtained at the start.

Participants were randomly assigned (1:1) to either the subtle or vibrant colour condition and viewed only one image. After exposure, they completed measures of DV, MC, CV, attention check and demographics.

Responses were excluded if participants failed the attention check that appeared in the middle of the survey or submitted incomplete surveys, to ensure quality data. The final sample consisted of N = 100 (50 per condition). The exported dataset included a binary condition variable, composite aesthetic appeal score, manipulation check, control variable, attention check indicator and demographic variables, structured for further analysis in R.

Data Analysis and Results:

Data was analysed in R to summarise descriptive statistics, assess internal consistency using Cronbach α, test manipulation check, evaluate balance across conditions and examine the main hypothesis using a two-sample t-test.

Below is a table that reports descriptive statistics by condition:

Condition	N	DV Mean (SD)	MC Mean (SD)	Age-Mean (SD)
Subtle	48	4.26 (0.87)	2.97 (0.89)	26.8 (6.10)
Vibrant	48	4.81 (0.95)	5.33 (0.73)	26.8 (5.26)

The CV was only checked for balance.

Internal consistency was acceptable for the DV (α = 0.83) and excellent for the MC (α = 0.90).

Overall, the two groups were well balanced in terms of participant age.

Please note that in all results, M_1 refers to the M_subtle and M_2 refers to M_vibrant.

The MC two-sample t-test implied that the vibrant condition was rated significantly more vivid than the subtle condition:
$M_1 = 2.97$, $M_2 = 5.33$, $t(90.63) = -14.22$, $p<0.001$

A gender balance check using the chi-square test proved that gender distribution did not differ significantly:
$\chi^2(2, N = 96) = 3.10, p = 0.21$

Interest in sports cars was also similar across conditions, indicating a balance in this aspect as well:
$M_1 = 1.94, M_2 = 2.00, t(93.98) = -0.39, p=0.695$

Consistent with H1, colour vibrancy significantly affected perceived aesthetic appeal. Participants rated the vibrant car as aesthetically more pleasing than the subtle car, with the following data on a two-sample t-test score.
$M_1 = 4.26, M_2 = 4.81, t(93.24) = -2.95, p=0.004$.

Figure 3, highlights the difference in mean across both conditions.

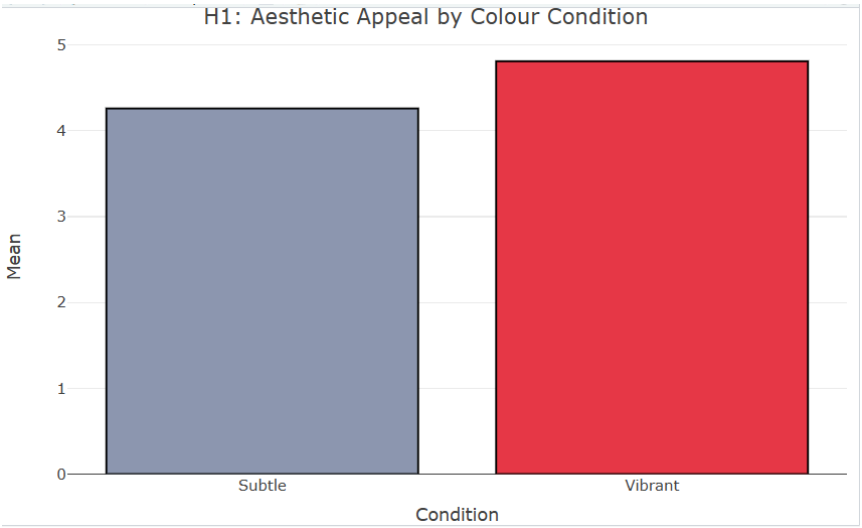


Figure 3: Barchart showing difference between DV's mean.

To conclude, the results support H1, showing that higher colour vibrancy leads to higher perceived aesthetic appeal, with manipulation working as intended and no imbalances between conditions.

Managerial Implications:

Across conditions, the high-vibrancy image received higher aesthetic appeal than the subtle version, suggesting vibrant colours improve immediate impressions in performance oriented categories. Vibrant variants must be prioritized in top-of-funnel visuals such as social media, thumbnails, launch teasers. Must be used cautiously where understated luxury/heritage cues dominate and test market fit and downstream outcomes such as CTR, willingness-to-pay.

Limitations and Further Research:

Some limitations include:

- The sample was AI-generated, so results may not reflect real consumer responses and must be interpreted cautiously
- Single brand/context (Porsche 911) limits generalisability across brands, categories and cultures.
- Only red was tested at two saturation levels, limiting relevance to other colours.

Further research should use real participants, larger samples, multiple brands/colours with outcomes such as purchase intent and willingness to pay to understand real buyer behaviour.

Reference list

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Appendix

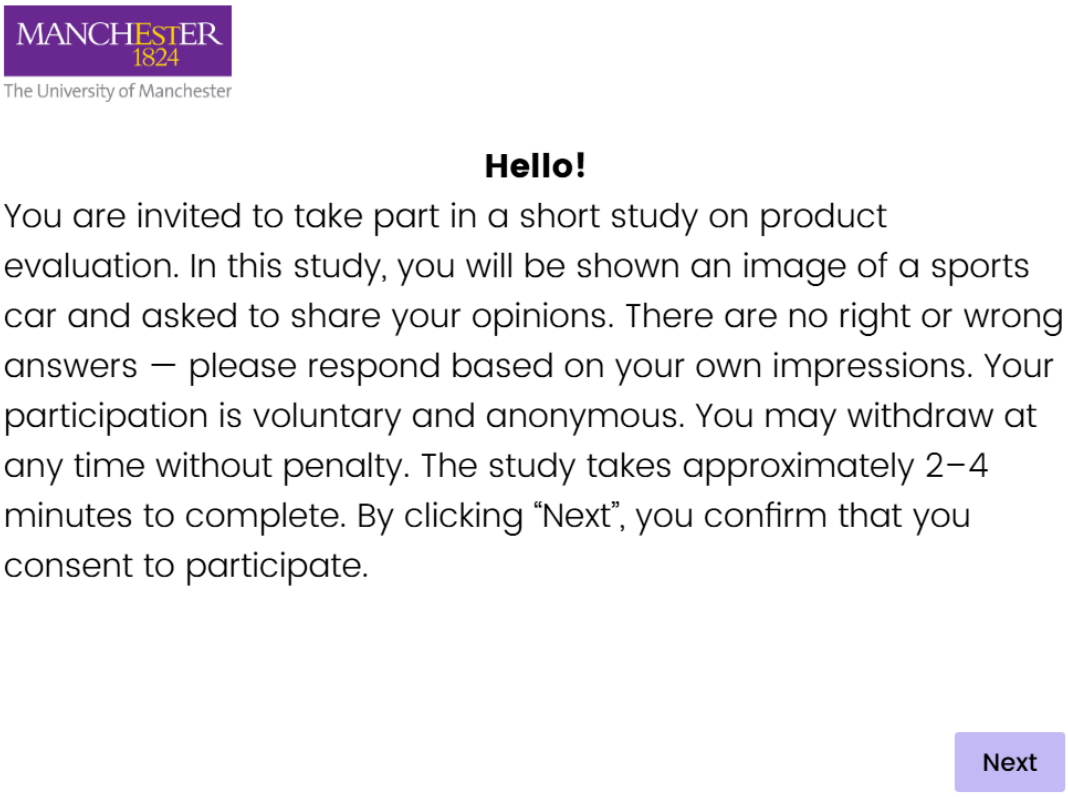
Appendix 1:

I had already put in my values for DV, MC, CV, Attention Check and Demographics earlier, the prompt used in ChatGPT to generate dataset is as follows:

Please read the attached document describing the experimental design and generate simulated survey responses. Simulate responses from 100 participants aged 18–35, with approximately 50 participants per condition. For each participant, generate responses for all variables measured in the survey, including the condition indicator, key outcome measures, manipulation check items, control variable items, and demographic variables. Ensure that the responses exhibit realistic variation across participants. Output the results in a table format, where each row represents one participant and each column represents one variable, using the variable names specified in the document. Generate a response for the Porsche 911 project. Additionally, Set 4 participants total to fail the attention check: 2 from Condition=0 and 2 from Condition=1. For these 4 participants, set AttentionCheck_5pt to any value other than 4 (e.g., 1, 2, 3, or 5). All other participants must have AttentionCheck_5pt = 4.

Appendix 2:

Photos of the questionnaire:



MANCHESTER 1824
The University of Manchester



The image on this page shows the design of a sports car. The brand name has been hidden for this study. Please carefully view the car below. You will be asked questions about it on the next page.

Next

MANCHESTER 1824
The University of Manchester

Based on the car you just viewed, please indicate how much you agree with the following statements

	Not at all	Mostly no	Slightly no	Unsure	Slightly yes	Mostly yes	Absolutely yes
This sports car is visually appealing.	○	○	○	○	○	○	○
I find the design of this car aesthetically pleasing.	○	○	○	○	○	○	○
I like the overall look of this car.	○	○	○	○	○	○	○

Based on the car you just viewed, please indicate how much you agree with the following statements

	Strongly disagree	Disagree	Slightly disagree	Neutral	Slightly agree	Agree	Strongly agree
The colour of this car appears vivid.	○	○	○	○	○	○	○
The colour of this car appears subtle.	○	○	○	○	○	○	○

When i see a sports car in general, i feel..

Unexcited	○	○	○	Excited
Indifferent	○	○	○	Enthusiastic
Uninterested	○	○	○	Very interested

To ensure data quality, please select "Somewhat agree" for this question.

- ○ Strongly Disagree
- ○ Somewhat disagree
- ○ Neither agree nor disagree
- ○ Somewhat agree
- ○ Strongly agree

Next



What is your age

What is your gender

- ○ Male
- ○ Female
- ○ Non-binary / third gender
- ○ Prefer not to say

Thank you for your participation. Your responses are anonymous and will be used for academic research purposes only.

Next

MANCHESTER 1824
The University of Manchester

We thank you for your time spent taking this survey.
Your response has been recorded.